“Energy Efficiency Optimization using IoT-Driven Power Consumption Analytics”

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

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Submitted by

S.Sai Varma (VU22CSEN0102367)
S.Manjith (VU22CSEN0101270)
Himanshu Singh(VU22CSEN0101246)
P.Lokesh (VU22CSEN0101012)

Under the esteemed guidance of
M N Vamsi Thalatam
Associate Professor

Department of Computer Science and Engineering/ Department of Artificial Intelligence and Data Science
GITAM School of CSE, GITAM (Deemed to be University), Visakhapatnam.

2025

DECLARATION

I hereby declare that the project report entitled **Smart Energy Management System using IoT and Machine Learning** is an original work done in the Department of Computer Science and Engineering /Department of Artificial Intelligence and Data Science, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering/ Computer Science and Engineering (AI&ML/DS/CS/IoT). The work has not been submitted to any other college or University for the award of any degree or diploma.

Date:24th November,2025

Registration No(s)	Name(s)	Signature
VU22CSEN0102367	S.Sai Varma	
VU22CSEN0101270	S.Manjith	
VU22CSEN0101246	Himanshu Singh	
VU22CSEN0101012	P.Lokesh	

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING/
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

GITAM School of CSE

GITAM (Deemed to be University)

CERTIFICATE

This is to certify that the project report entitled "**Smart Energy Management System using IoT and Machine Learning**" is a bonafide record of work carried out by S.SaiVarma (VU22CSEN0102367), S.Manjith (VU22CSEN0101270), Himanshu Singh (VU22CSEN0101246),P.Lokesh (VU22CSEN0101012) students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering / Computer Science and Engineering (AI&ML/DS/CS/IoT).

Date :24th November 2025

Project Guide : M N Vamsi Thalatam Head of the Department

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Abstract

This project presents a Smart Energy Management System designed to optimize power consumption using Internet of Things (IoT) sensors and Machine Learning (ML) models. The system collects real-time parameters such as voltage, current, power, and energy usage from IoT-enabled devices through ESP32-based hardware. The data is processed and analyzed using clustering algorithms to identify inefficient usage patterns and deep-learning models such as LSTM to forecast future power demand. A user-friendly dashboard displays consumption trends, alerts, and recommendations. The proposed system aims to reduce energy wastage, improve efficiency, and support sustainable practices in residential and industrial environments.

Introduction

Electricity consumption has become an integral part of modern life, powering homes, industries, transportation, and communication systems. With increasing urbanization and the proliferation of electronic devices, energy demands have risen sharply, often leading to wastage due to:

- Improper appliance usage
- Inefficient load management
- Lack of awareness of real-time consumption
- Absence of predictive tools for load forecasting
- Traditional meters showing only cumulative readings

Smart Energy Management Systems (SEMS) aim to bridge this gap by leveraging the combined power of **Internet of Things (IoT)** and **Machine Learning (ML)** technologies. IoT sensors collect granular data in real time, while ML models interpret patterns and make future predictions.

The key goals of SEMS include:

- Monitoring electrical parameters in real-time
- Analyzing energy consumption behaviour
- Identifying inefficiencies proactively
- Predicting future load to avoid peak-time wastage
- Providing actionable recommendations to users

This project applies IoT and ML techniques to design an intelligent system that provides users with real-time and predictive insights on energy consumption, empowering them to make informed decisions and adopt sustainable energy practices.

Literature Review

These papers highlight the key advancements in IoT energy monitoring and ML-based load forecasting, and collectively form the core findings that guided the development of our proposed system.

Paper 1: IoT-Based Smart Metering

This study demonstrated the effectiveness of ESP32 and PZEM-004T sensors for real-time monitoring of voltage, current, and power. It proved that IoT hardware enables accurate, low-cost data collection, but the system lacked predictive analytics and behaviour detection, highlighting the need for ML integration.

Paper 2: LSTM for Energy Forecasting

Research applying LSTM networks to smart meter data showed that LSTM significantly outperforms traditional models like ARIMA due to its ability to capture long-term temporal dependencies. However, these studies focused solely on forecasting and did not incorporate IoT data or behaviour analytics.

Paper 3: Clustering for Consumption Behaviour

A study using K-Means clustering on household energy data revealed that clustering can effectively categorize users into groups such as high-load or irregular-usage consumers. While insightful, the work lacked forecasting capability and real-time sensor integration.

Paper 4: Machine Learning on Smart Meter Data

Researchers used Random Forest and XGBoost to identify major energy-consuming appliances. Although the models gave high accuracy, the study was limited to offline analysis and did not combine real-time IoT monitoring with predictive modelling.

Paper 5: IoT-Assisted Home Energy Optimization

This work used IoT sensors for automating appliance control based on environmental conditions. It reduced energy wastage but focused more on automation than on pattern

analysis or forecasting, leaving room for integrated IoT-ML solutions like the one proposed in this project.

Paper 6: IoT and Machine Learning for Predictive Maintenance

This study explored the integration of IoT sensors with machine learning models to detect abnormal energy consumption and predict equipment faults in industrial environments. The system improved efficiency and reduced energy loss, but it was primarily designed for predictive maintenance rather than household-level consumption analysis. It also lacked behavioural clustering and load forecasting, indicating the need for a broader, consumer-oriented SEMS like the one proposed in this project.

Problem Identification

Despite advancements in energy metering technology, consumers still lack detailed insights into their electricity usage behaviour. Traditional power meters present cumulative energy readings but fail to provide real-time monitoring, behaviour analysis, abnormal consumption detection, or predictive insights. Users are unable to identify which devices or usage patterns contribute most to energy wastage. Additionally, the absence of forecasting means that consumers cannot anticipate future spikes in energy consumption, leading to unplanned overuse and increased electricity bills.

These limitations highlight the need for an intelligent, automated system capable of transforming raw electrical data into meaningful information. The problem is therefore centered around the challenge of developing a system that not only measures consumption but also interprets and predicts it, enabling users to proactively manage their energy usage.

Objectives

Phase 1 – Machine Learning Model Development

- Preprocess UCI dataset
- Build clustering model to identify usage profiles
- Train LSTM to forecast next-hour power usage
- Prepare dashboard-ready outputs

Phase 2 – IoT Integration

- Build ESP32-PZEM based smart meter
- Stream real-time data to backend
- Run ML inference on IoT data
- Provide real-time recommendations via dashboard

Existing System

Existing electricity monitoring systems typically rely on analog meters or digital meters that show only cumulative consumption. These systems lack the capability to monitor instantaneous power, identify patterns, detect inefficiencies, or predict future load. Moreover, users are unable to view detailed graphs or receive notifications regarding abnormal behaviour or peak usage times.

Even advanced smart meters provided by utility companies mainly focus on billing purposes rather than user empowerment. They rarely offer analytics, behaviour insights, or predictive statistics. In contrast, modern IoT and ML technologies have the potential to transform energy monitoring into a proactive, intelligent process. However, the absence of integrated solutions combining real-time IoT data with ML-driven insights remains a major gap in the industry.

PROPOSED SYSTEM

The proposed Smart Energy Management System addresses the limitations of existing solutions by integrating IoT-based sensing, machine learning analytics, and visualization tools into a single, cohesive platform. The system enables real-time acquisition of electrical parameters through an ESP32 microcontroller connected with a PZEM-004T energy sensor. These readings are forwarded to the backend for processing.

At the processing layer, the raw data undergoes cleaning, normalization, and feature extraction before being fed into two machine learning modules. The **clustering module** groups consumption patterns into categories to highlight inefficiencies. Meanwhile, the **LSTM forecasting module** predicts upcoming electricity demand, enabling proactive decision-making.

The results are displayed via an interactive dashboard that provides live monitoring, historical trends, alerts for abnormal behaviour, and personalized energy-saving recommendations. This system ensures that users gain complete visibility into their energy consumption while benefiting from the predictive power of ML techniques.

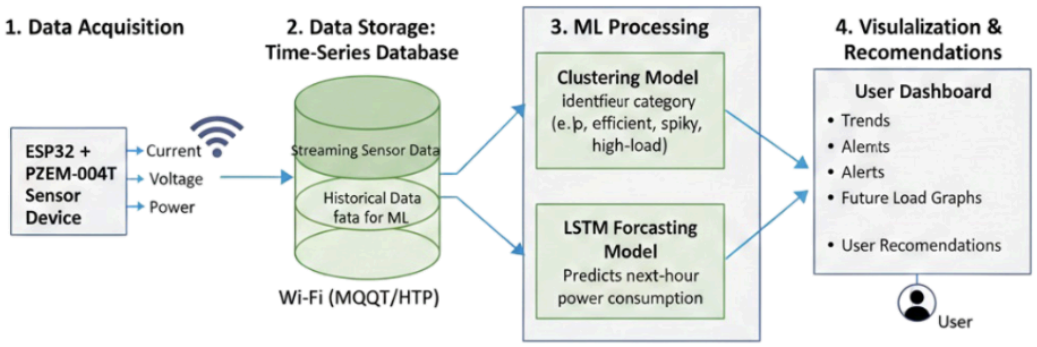

Figure 1. Proposed Architecture

SYSTEM ARCHITECTURE

The system architecture of the Smart Energy Management System (SEMS) is designed as a multilayered framework that integrates IoT-based data acquisition, backend data processing, machine learning analytics, and user-level visualization. The architecture follows a modular design to enhance scalability, interoperability, and deployment flexibility.

At the lowest layer lies the IoT Sensing Layer, which consists of the ESP32 microcontroller interfaced with the PZEM-004T energy sensor. This layer performs real-time measurement of electrical parameters, including voltage, current, active power, and energy consumption. These readings are streamed at fixed intervals to the processing layer.

The Communication Layer establishes reliable data transmission between the physical smart meter and the cloud or local server. The ESP32 communicates via Wi-Fi using HTTP or MQTT protocols. This layer ensures secure, low-latency transfer for real-time analytics.

The Data Processing Layer handles tasks such as data cleaning, noise removal, timestamp correction, resampling, normalization, and feature engineering. It converts raw IoT readings into structured, ML-compatible datasets.

Above this lies the Machine Learning Layer, which houses two independent models: a clustering model and a forecasting model. The clustering model classifies consumption behaviour into patterns such as “efficient,” “high-load,” or “irregular usage,” while the LSTM model predicts the next-hour consumption value based on historic trends.

Finally, the Application Layer includes the dashboard interface responsible for presenting real-time data streams, ML predictions, alerts, historical trends, and personalized recommendations to the user. This layered architecture ensures smooth end-to-end functionality, from data sensing to actionable insight generation.

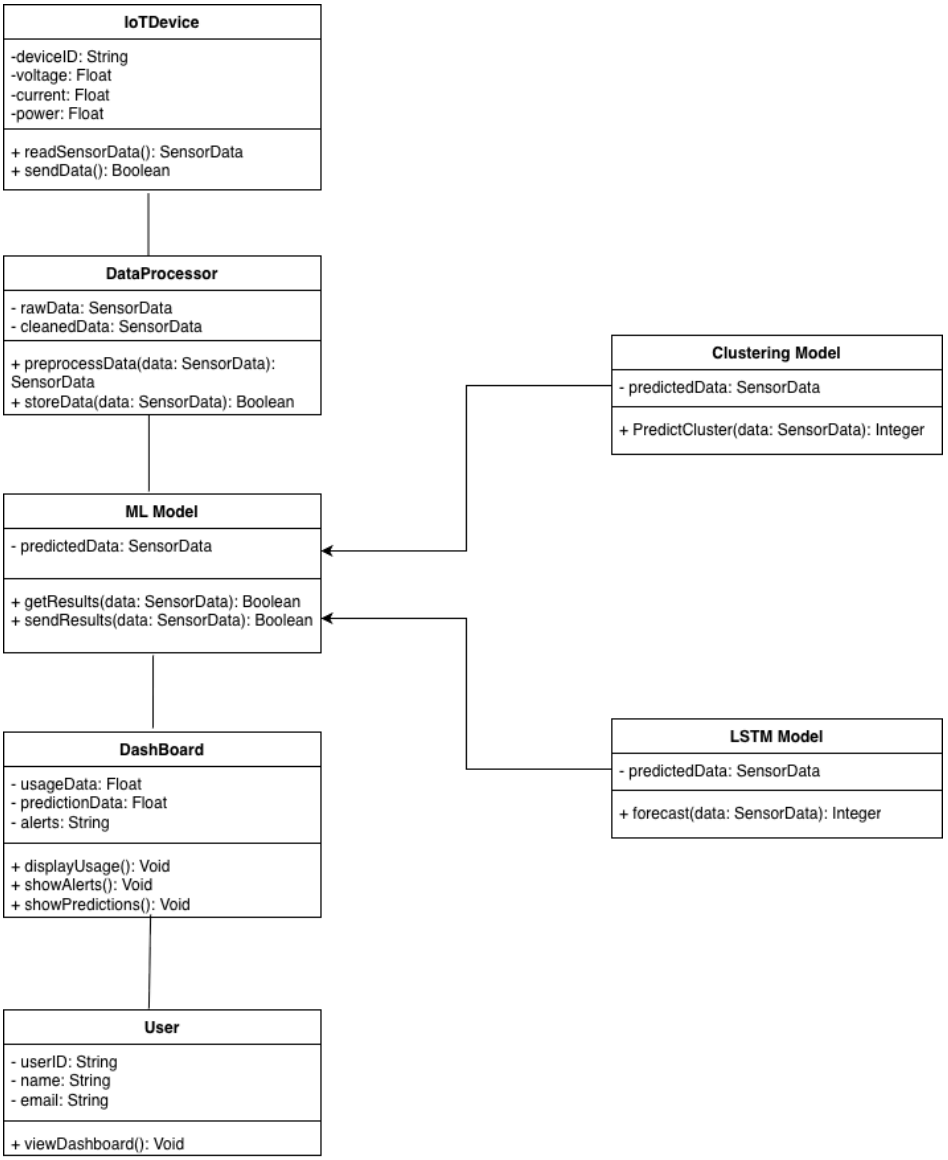

Figure 2.UML Class Diagram

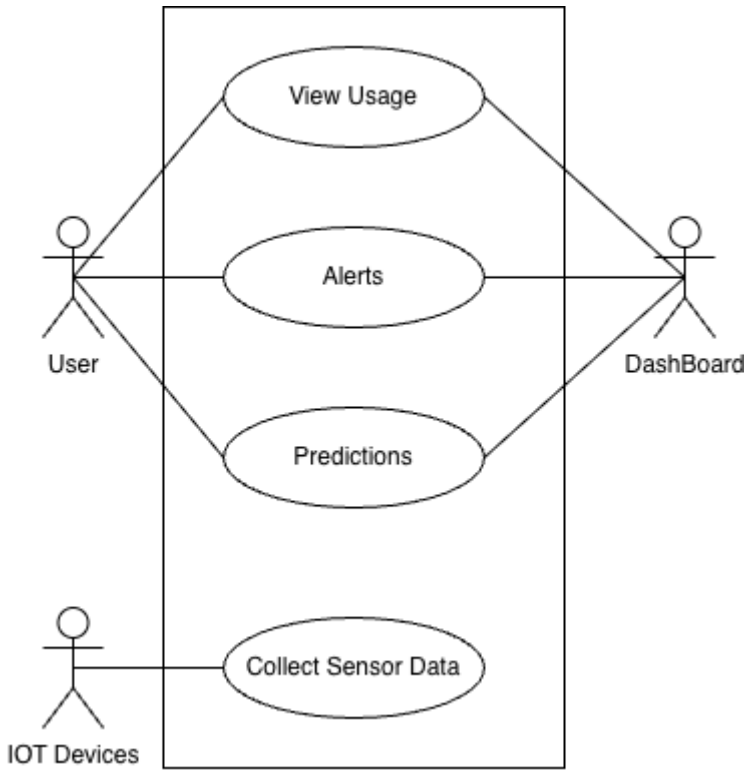

Figure 3.UML UseCase Diagram

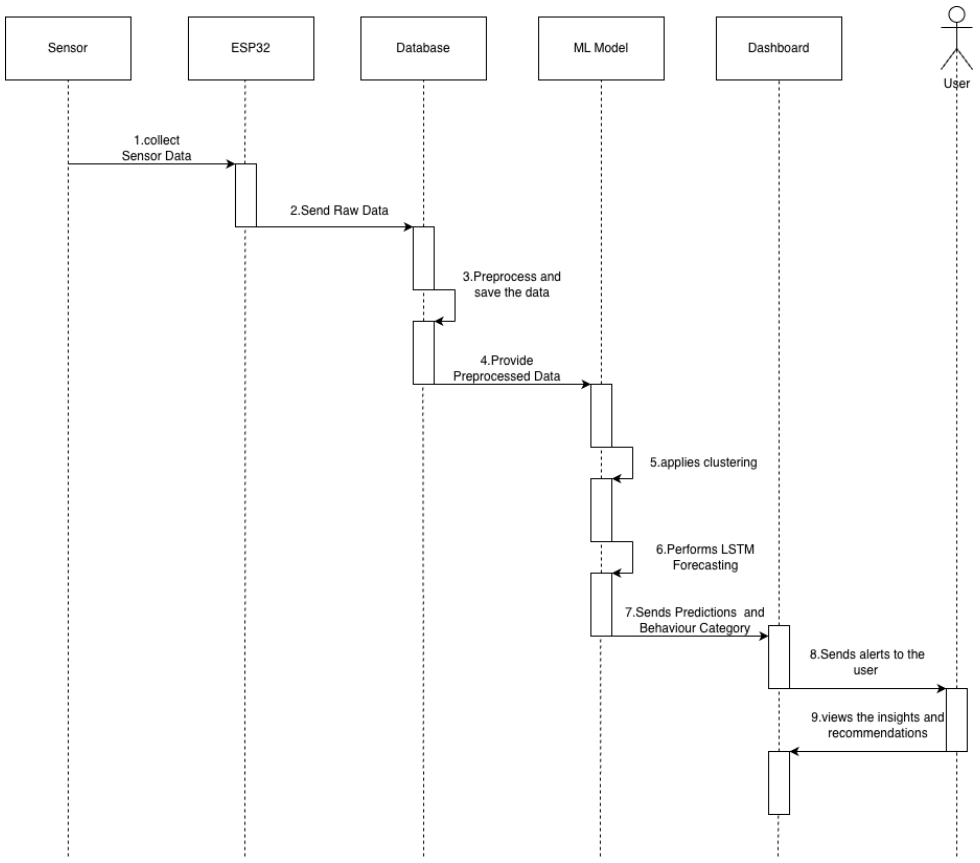

Figure 4.UML Sequence Diagram

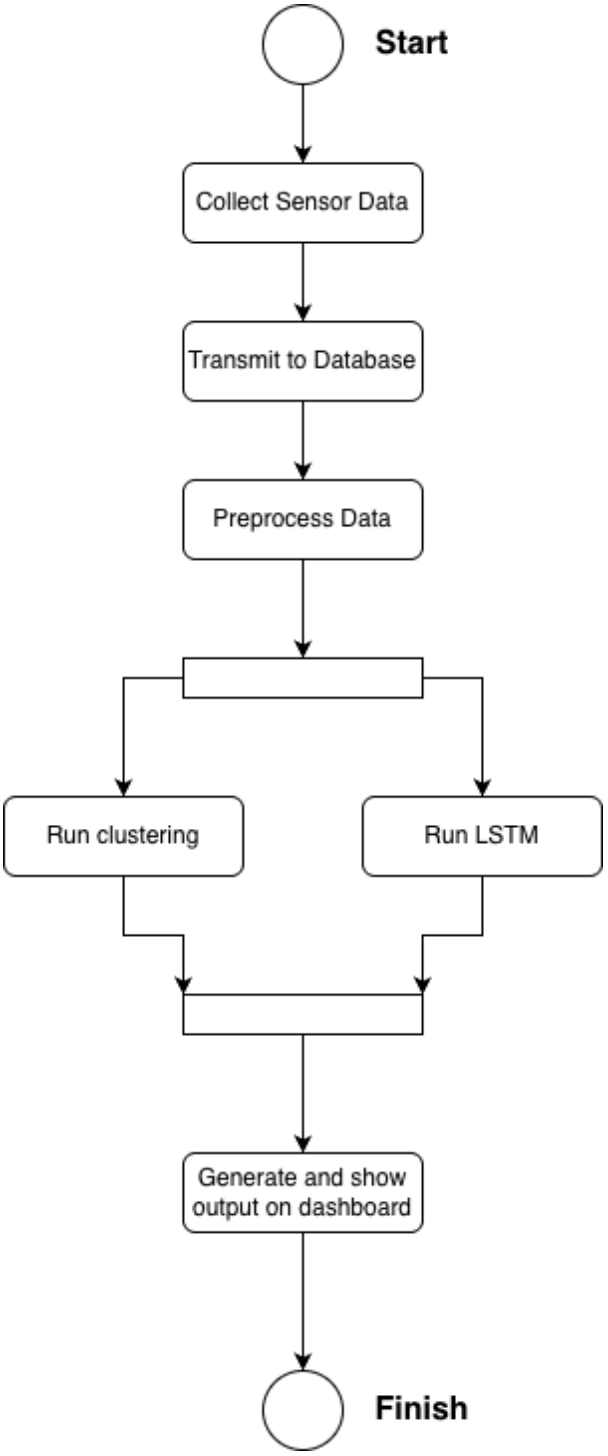

Figure 5.UML Activity Diagram

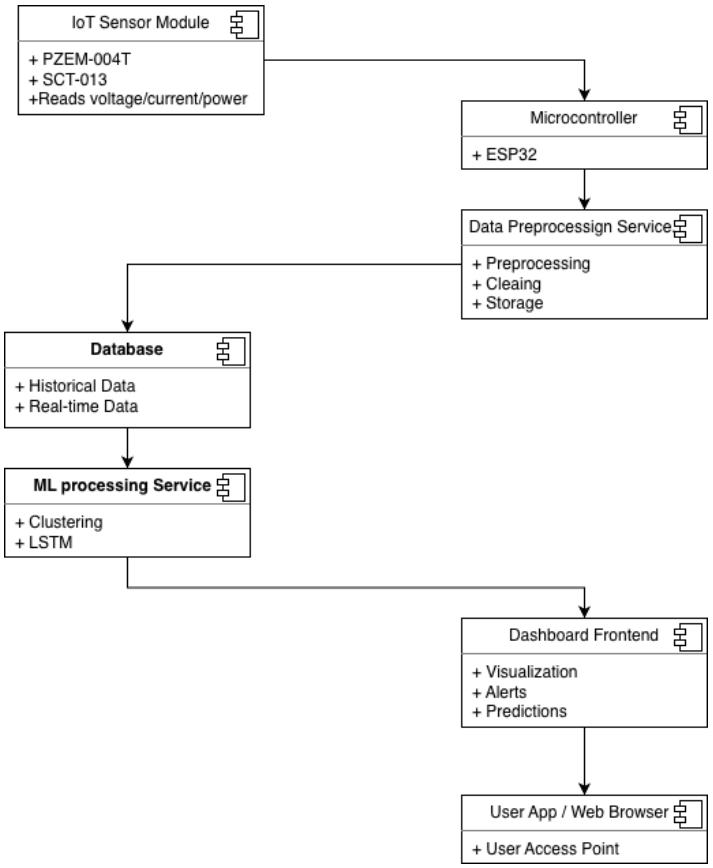

Figure 6.UML Component Diagram

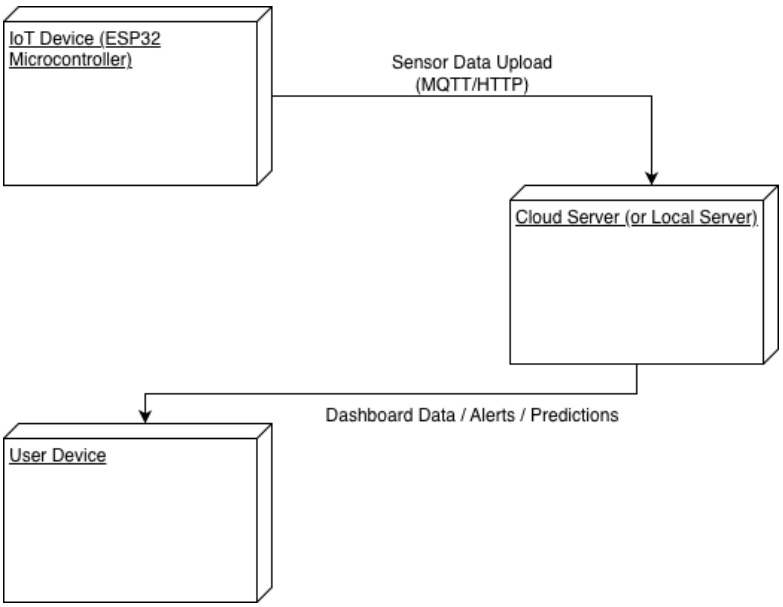

Figure 7.UML Deployment Diagram

Tools/Technologies Used

The system integrates a variety of hardware and software technologies to support its end-to-end operation.

Hardware Technologies

- **ESP32 Microcontroller:** A powerful Wi-Fi-enabled microcontroller used for real-time sensor interfacing.
- **PZEM-004T v3 Energy Sensor:** Measures voltage, current, active power, and energy consumption with high accuracy.
- **SCT-013 Current Transformer (optional):** Supports current measurement in extended configurations.

Software Technologies

- **Python:** Primary programming language for data processing and machine learning model development.
- **Pandas & NumPy:** Used for data manipulation, preprocessing, and mathematical operations.
- **TensorFlow/Keras:** Framework for designing and training the LSTM forecasting model.
- **Scikit-Learn:** Used for clustering, scaling, and evaluating machine learning models.
- **Streamlit/React:** For dashboard development and real-time visualization.
- **Firebase:** Backend storage systems for live IoT data.
- **MQTT/HTTP:** Communication protocols for sending IoT data to backend.

Algorithms

- **Clustering (K-Means):** K-Means groups energy consumption data into clusters based on similarity, helping identify different usage behaviour patterns.
- **LSTM Forecasting:** LSTM predicts future energy consumption by learning patterns from past time-series data.

Conclusion

The Smart Energy Management System presented in this project provides an integrated solution for monitoring, analyzing, and optimizing electricity consumption. By combining IoT hardware with machine learning techniques, the system dynamically captures real-time energy data and converts it into actionable insights. Clustering models identify inefficient consumption behaviours, while LSTM models provide accurate predictions of future demand. The dashboard acts as a user-centric gateway to visualize patterns and adopt energy-efficient practices. This project demonstrates the potential of IoT-ML integration in achieving sustainable energy usage and paves the way for intelligent home automation and smart grid applications.

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